Help Desk: First Shift

A CTF for Cisco NetAcad **IT Customer Support Basics** (Modules 1–2) · Facilitator guide & answer key · **Instructor only**

> **At a glance:** 33 challenges · 7,550 points · 8 categories that follow course topics 1.1–2.4 · 16 evidence files · tested on CTFd 3.8.6 · about 90–120 minutes · individual play · students need a spreadsheet app and a text editor, and internet for one challenge (Stop Code Detective).

The story

Students are the newest Tier 1 technicians at **Sawgrass Logistics**, a fictional company. They work a real-looking ticket queue, review call transcripts and ticket notes, read remote-support audit logs, and check AI output and research sources. Most flags come from **analyzing evidence**, not recalling facts. Only four challenges are straight vocabulary (warm-ups). The locked boss ticket makes students chain three files, which is what a real escalation looks like.

Setup

1. Set up CTFd and finish the setup wizard. See `docs/platform-setup.md` in the OmniTech-Arena repository.
2. In CTFd as admin: **Settings** → **Access Tokens** → **Generate**.
3. From the repository root: `pip install requests`, then `python tools/deploy.py tracks/helpdesk-cisco-it-support/01-first-shift_it-customer-support-basics --url http://<host>:8000 --token ctfd_...`
4. Look for `Done: 33 challenges loaded.` Run it **once**. Other games aren't touched. Add `--hidden` to load the game hidden.
5. One time per server: paste the two sort functions from `docs/platform-setup.md` into **Admin Panel** → **Config** → **Theme** so each game's categories stay grouped.

Running the event

Time	What happens
0:00–0:10	Students register (tell them to use first name + last initial). Show the board. Everyone solves **Clock In**. Remind them that answers aren't case-sensitive and that some challenges have limited attempts, shown under the box.
0:10–1:30	Open play. Students choose their path. Beginners do well starting with categories 02, 03 and 04. Your spreadsheet-strong students will go straight for Breach! and First Contact.
~0:45	Check the scoreboard. If many students are stuck on Breach!, do a two-minute mini-lesson on subtracting timestamps in Excel: `=(resolved-created)*24`.
1:30–2:00	Debrief. The best discussion prompts are in the answer key: *Robot Coworker* (trusting AI), *Hold, Please* (role-play lines 06/13/18 the right way), *Trust but Verify*, and the boss ticket (why good session notes and KB articles saved the day).

Hints: Free hints (0 points) nudge toward the method. Paid hints cost 20–50 points and give more away. To make it easier, set all hint costs to 0 in the Admin panel. To make it harder, delete the free hints.

Curriculum alignment

Each challenge is tagged in CTFd with its CCST IT Support objective (1.1, 1.2, 1.3, 4.5, 6.1, 6.2) and, where it fits, a Y100200 standard: 02.0 troubleshooting, 03.0 OS & software, 04.0 networking, 05.0 security threats, 07.0 operational procedures, 09.0 math, 13.0 communication, 14.0 problem solving, 15.0 IT tools, 17.0 ethics & legal. Tags show on each challenge card, so students can see what each one practices.

Answer key

Flags accept reasonable formatting: `INC-10408` or `10408`, `1,3,5,8` or `1 3 5 8`, `DEACB` or `D-E-A-C-B`. The order still matters. Every answer below was submitted by a test student account on a live CTFd 3.8.6 install and accepted, and several wrong answers were checked and rejected.

Challenge	Answer	Why / discussion	Files
01 Orientation (1.1)			
Clock In 50 pts	`first shift`	Teaches the answer box. Everyone should get this in the first minute.	–
Alphabet Soup 100 pts	`Service Level Agreement`	Sets expectations for response and resolution times (1.1).	–
Set in Stone 100 pts	`ticket number (ticket ID also accepted)`	The unique, auto-generated identifier that never changes.	–
KPI Speedrun 100 pts	`First Contact Resolution (FCR, first call resolution also accepted)`	KPI named in the Module 1 new-terms list.	–
Queue Jumper 200 pts 5 tries	`INC-10408`	Two New P1 tickets: INC-10408 (dispatch floor, 07:18) and INC-10417 (warehouse scanners, 08:03). The oldest one wins. INC-10412 is also P1, but it's already In Progress.	ticket_queue_export.csv
Breach! 300 pts 6 tries	`5`	Breaching tickets: INC-10383, INC-10387, INC-10392, INC-10396, INC-10401. Resolution hours compared with 4/8/24/72.	ticket_queue_export.csv sla_policy.txt
First Contact 300 pts 6 tries	`64`	16 of 25 resolved tickets have contacts = 1. 16/25 = 64%.	ticket_queue_export.csv sla_policy.txt
Fast Response 250 pts 6 tries	`12`	P1 first-response minutes: 8, 12, 22, 5, 13. Average 12. Check this against the 15-minute target as a discussion point.	ticket_queue_export.csv
Same Problem, Different People 200 pts 5 tries	`CPY-2F-01`	Tickets INC-10406, INC-10409, INC-10413, INC-10418 describe the same copier four different ways. Group them and work one root cause. KB0142 covers SC-542.	ticket_queue_export.csv

Robot Coworker 250 pts 4 tries	`INC-10414`	Grace Liu's account lockout was AI-categorized as Hardware. It should be Account/Access, and it's arguably P2. This is a good talking point about checking AI output.	ticket_queue_export.csv
02 Front Line (1.2)			
Open or Closed? 150 pts 5 tries	`1,3,5,8`	Numbers 2, 4, 6 and 7 can be answered yes/no or by picking one option. Any separator is accepted.	–
Who's On the Line? 300 pts 4 tries	`DEACB`	Talkative = D (wedding story), Rude = E (insults, wants Marcus), Knowledgeable = A (already flushed DNS), Angry = C (lost work, long hold), Inexperienced = B ("the blue e").	call_transcripts.txt
Hold, Please 250 pts 4 tries	`6,13,18`	06 holds without asking or giving a time. 13 transfers without explaining why or to whom. 18 is a cold bounce back to the queue that drops the call. 09 is the model of doing it right, and 20 shows recovery and ownership.	hold_and_transfer_call.txt
Say It Back 200 pts 2 tries	`C`	C paraphrases accurately in plain language and confirms. A is jargon, B is too vague, and D skips listening. Limited to 2 attempts.	–
03 Paper Trail (1.3)			
Loose Lips 200 pts 3 tries	`INC-10303`	The user's password is written into the ticket note, which breaks confidentiality.	ticket_notes_review.txt
Just the Facts 200 pts 3 tries	`INC-10304`	Personal opinion and a stereotype about the user's department.	ticket_notes_review.txt
Unfinished Business 250 pts 3 tries	`INC-10305`	Steps are listed, but there's no Outcome or Next steps.	ticket_notes_review.txt
Deja Vu 300 pts 4 tries	`31.0.15.4601`	Step 4 of INC-09811. This is the course's Diego blue-screen scenario, extended to show why documentation matters.	diego_ticket_history.txt
04 Troubleshooting (2.1)			
Eight Steps 200 pts 5 tries	`BEHDGAFC`	Define, Gather, Probable cause, Plan, Implement, Observe, Repeat, Document.	–
One Thing at a Time 250 pts 2 tries	`6`	Entry 6 makes four changes at once, which is why entry 8 can't explain the fix. Kyle also never documented anything useful (a discussion point).	troubleshooting_log_kyle.txt
What Changed? 300 pts 4 tries	`23.60.0.9`	Installed 9/26 16:22, followed by 'requires further installation' and then 'adapter not present' on Monday. The Defender update came after the failure, and 23.40.0.4 worked for weeks. KB0217 has the fix.	room114_laptop_events.txt
05 Remote Access (2.2)			
Right Tool, Right Job 250 pts 4 tries	`BDCEA`	1 Remote Assistance, 2 macOS Remote Desktop, 3 Webex, 4 TeamViewer, 5 Microsoft Remote Desktop.	–
Who Let You In? 250 pts 4 tries	`RS-4479`	VNC session connected with 'admin override - user prompt skipped'. There's no USER_ACCESS_GRANTED.	remote_sessions.log

Close the Door 250 pts 4 tries	`RS-4483`	It never logs DISCONNECTED or LOGGED_OUT, and the tech 'left for break'. RS-4483 also accepted without the prefix.	remote_sessions.log
06 Research Desk (2.3)			
Stop Code Detective 200 pts 5 tries	`DRIVER_POWER_STATE_FAILURE`	Bug check 0x9F (Microsoft Learn). Needs internet access. Spaces instead of underscores are accepted.	–
Trust but Verify 300 pts 3 tries	`/resetstack`	Not a real ipconfig switch. The real reset is `netsh int ip reset`. The other five commands are real. Students can check with `ipconfig /?`.	ai_chat_export.txt
Prompt Check 250 pts 3 tries	`2,4`	2 leaks an email, a password and an employee ID into a public AI. 4 has role, context, the specific error, the format and the ordering.	ai_prompts.txt
Source Check 200 pts 2 tries	`C,B`	Try C first (Microsoft, current). Avoid B (unknown .exe that says to disable antivirus). A is outdated (Outlook 2010 / Windows 7) and deletes a whole registry key. D is useful but secondary.	research_results.txt
Check the KB First 200 pts 4 tries	`KB0317`	Matches caa70004 after a password change.	internal_kb_index.txt
07 App Support (2.4)			
Where'd You Get That? 200 pts 4 tries	`PDFEditorPro_FREE_Setup.exe`	Downloaded from an unknown website rather than the Store or Company Portal.	app_install_audit.log
Why Won't It Install? 250 pts 3 tries	`CIPC`	1 needs Windows 11 (C), 2 corrupt download after a network change (I), 3 blocked by the org (P), 4 device not supported (C).	store_install_errors.txt
Mail Call 250 pts 3 tries	`CFG,SYN,NET`	1 wrong SMTP server and port, 2 desktop not syncing while the network is fine, 3 hotel captive portal not accepted.	email_tickets.txt
08 Boss Ticket			
The CFO's Month-End Meltdown 500 pts 3 tries	`RC-OFC142`	INC-10412 (CFO) leads to session RS-4480, which notes 0xc0000142 and 'look up by error code'. That points to KB0288, resolution code RC-OFC142. It unlocks after Queue Jumper, Close the Door and Check the KB First.	ticket_queue_export.csv remote_sessions.log internal_kb_index.txt

Extending it

- **Write-up round (not auto-graded):** after the boss, have students write the full ticket note for INC-10412 using the Module 1.3 format (issue, steps, outcome, next steps, no sensitive data). Grade it in Canvas.
- **Role-play round:** pairs act out calls C and E from `call_transcripts.txt`, then redo `hold_and_transfer_call.txt` lines 06, 13 and 18 correctly.
- **Reuse:** `challenges.json` is plain text. Change the answers in the evidence files and flags to make a new version for each semester.